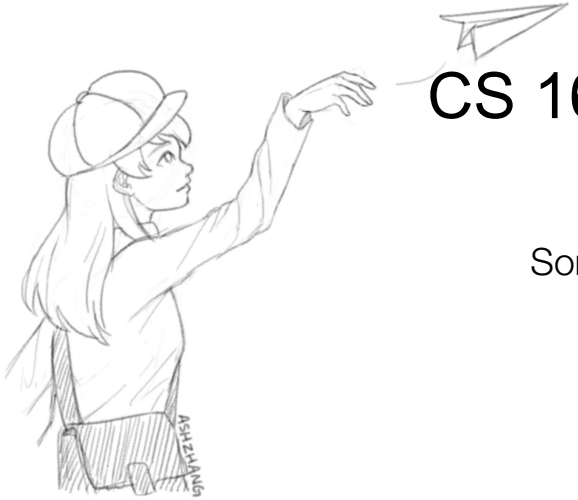
Note to self:
Set up terminal

Introduction to Networking



CS 161 Spring 2026 - Lecture 23

Some slides (as marked) are courtesy of UCB graduate and security expert Grant Ho.

Why study network protocols in a security class?

- Most people do not have **physical access** to your computer
  - Exceptions are few:
    Partner at home, law enforcement after stop, laptop kidnapping, …

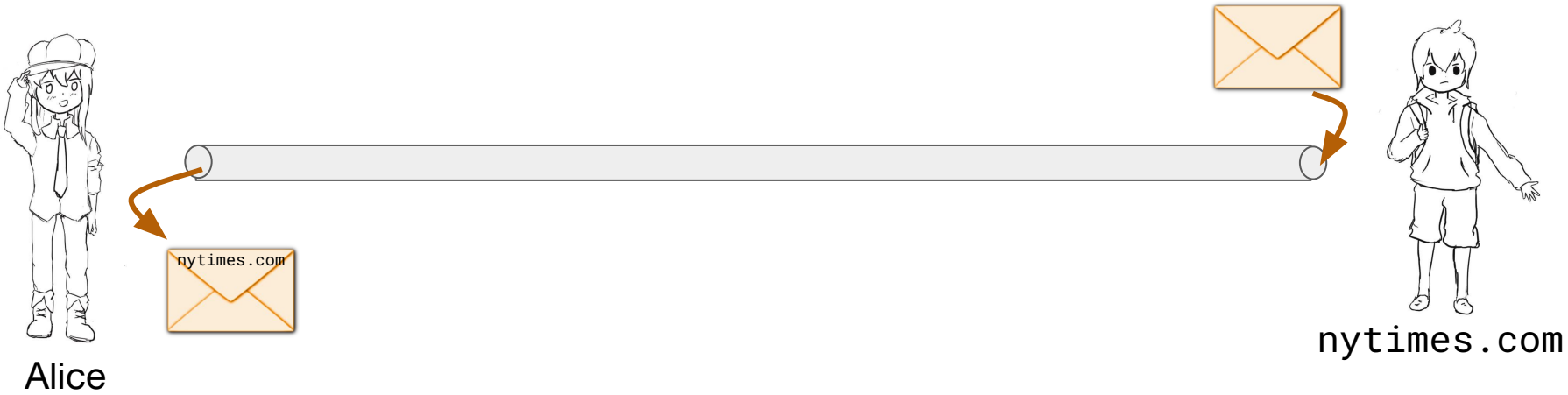
Important advice from UCPD: "If you must sleep, sleep on your computer."

- In contrast, literally **EVERYONE ON THE INTERN**
  has **network access** to your Internet-connected computer
  - Criminal outfits who want your money (ransomware) or other people's money (spam)
  - Sanctioned governments who want your bitcoin
  - Competitors who want your company's sweet sweet language models (or code)
  - …

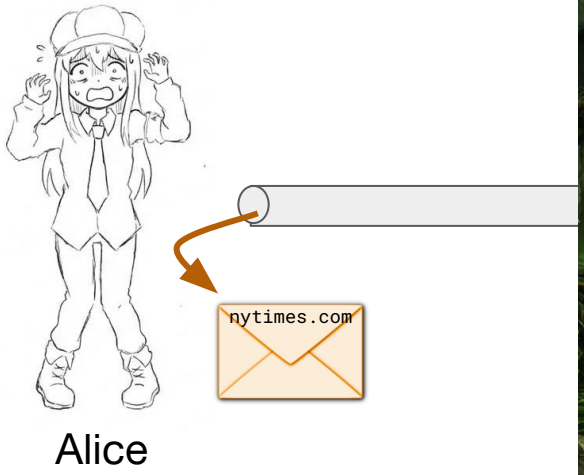
Fortunately, nothing important is connected to the Internet!

- Your bank
- Your car
- The database storing your medical records
- Your airplane (if you got into AI at the right time)
- …

How you think about the internet before CS161



How you think about the internet after CS161



Alice



The bottom line

- The “classic” suite of Internet protocols (TCP, UDP, Ethernet, …) provides
  - **NO** confidentiality
  - **NO** authentication / integrity protection

- What if you really want confidentiality/integrity protection?
  - Cryptographic protocols: TLS, encrypted messaging, etc.

- Cryptography can’t always save you!
  - Denial of service
  - Traffic analysis
  - …

Finnish police seize ship suspected of sabotaging undersea cable

31 December 2025

Share Save

Gabriela Pomeroy



Gulf of Finland Coast Guard (SLMV) / Handout/Anadolu via Getty Images

The Finnish coastguard approached the vessel and detained its crew

Finnish police have detained a vessel sailing from Russia, which is suspected of damaging an undersea telecoms cable running from Helsinki to Estonia across the Gulf of Finland.

https://www.bbc.com/news/articles/c62040np372o

Today: Intro to Networking

- The Internets: What is it???
- OSI model and layers
- Common Internet protocols

What's the Internet?

Textbook Chapter 25

What’s the Internet?

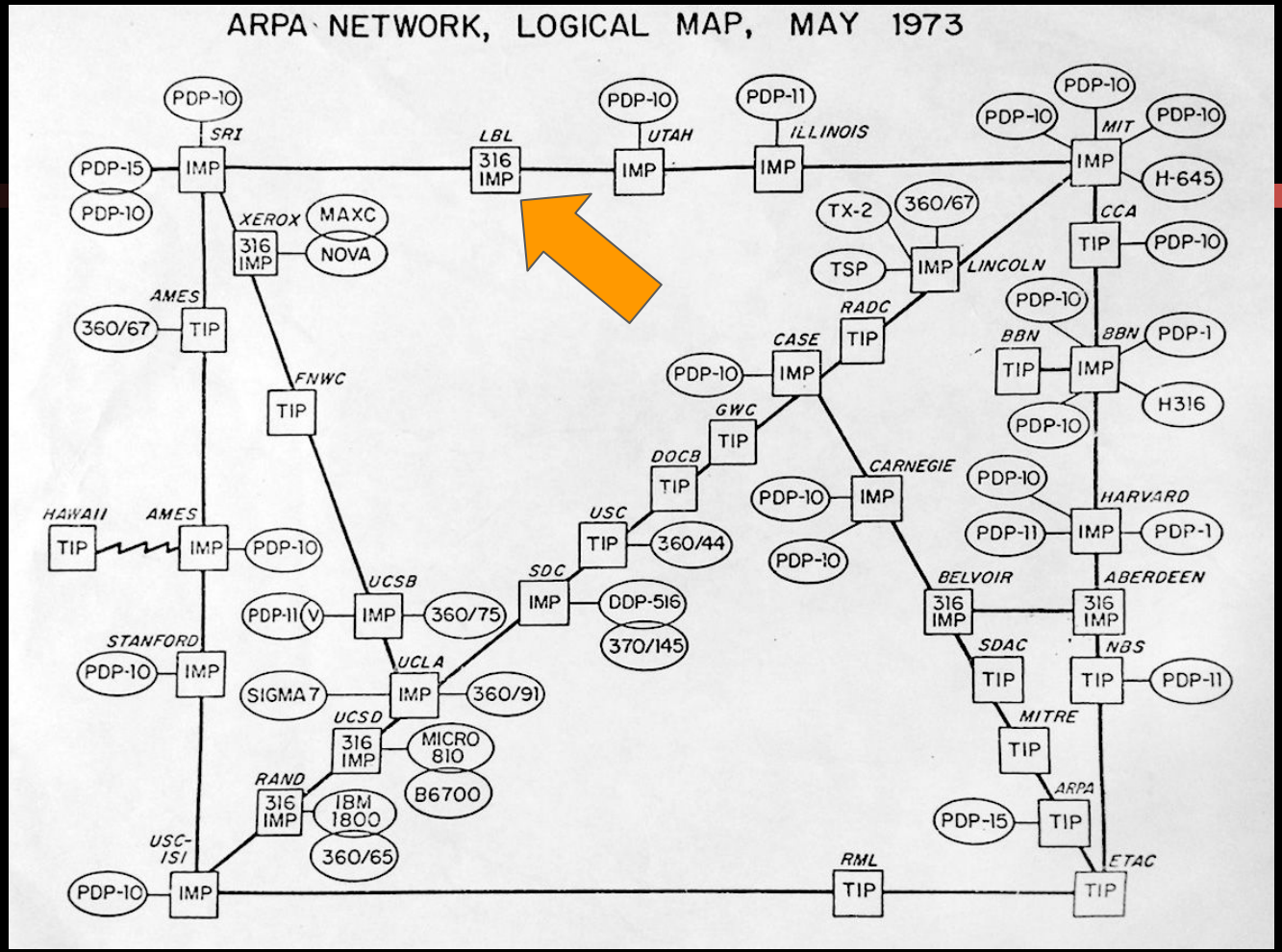
- **Network**: A set of connected machines that can communicate with each other
  - Machines on the network agree on a set of **protocols**, a set of rules for communication

- **Internet**: A global network of computers
  - The web sends data between browsers and servers using the Internet
  - The Internet can be used for more than the web (e.g., SSH)

Protocols

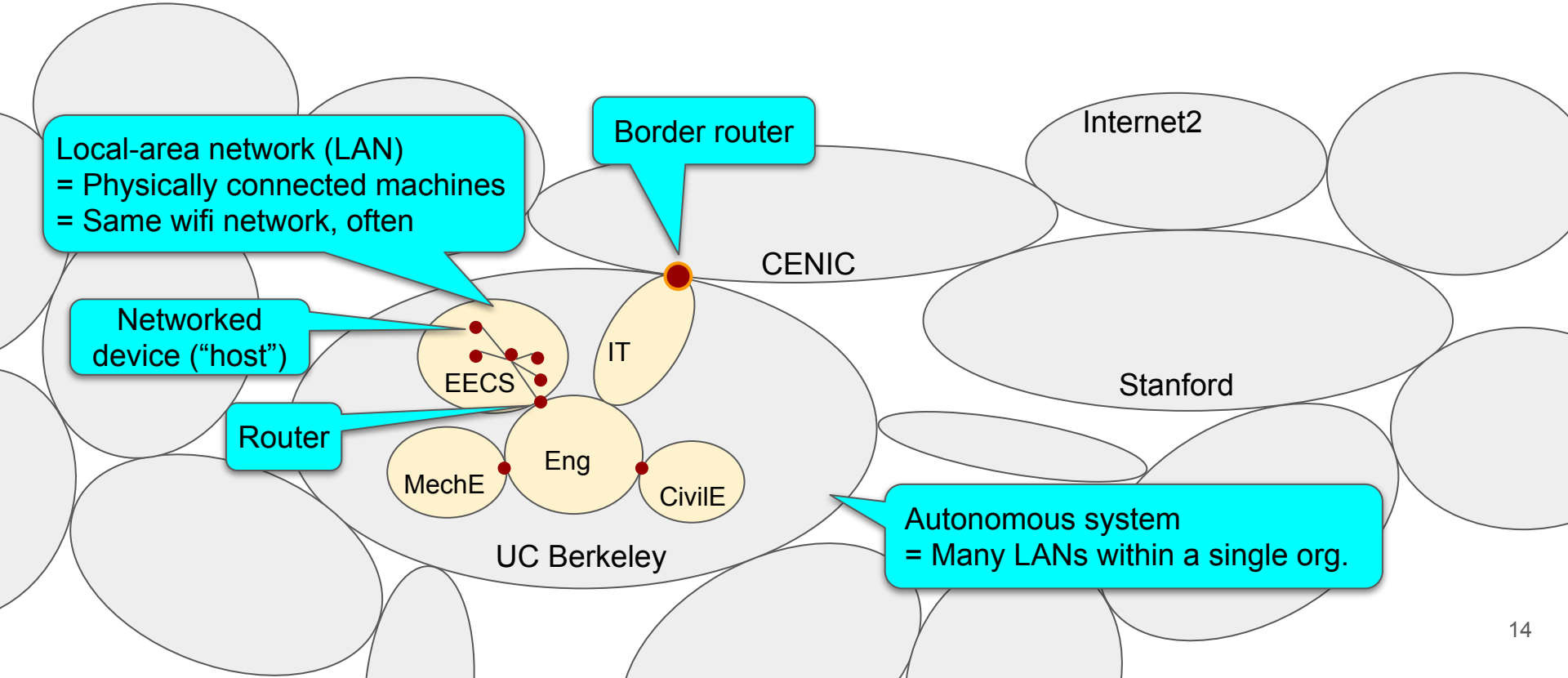
- A **protocol** is an agreement on how to communicate, specifying syntax and semantics
  - *Syntax:* How a communication is specified and structured (format, order of messages)
  - *Semantics*: What a communication means (actions taken when sending/receiving messages)

- Example: Protocol for asking a question in lecture?
  1. The student should raise their hand
  2. The student should wait to be called on by the speaker or wait for the speaker to pause
  3. The student should speak the question after being called on or after waiting
  4. If the student has been unrecognized after some time: Vocalize with “Excuse me!”



https://upload.wikimedia.org/wikipedia/commons/b/bc/Arpanet_map_1973.jpg

Cartoon version of the Internet





https://maps.cenic.org/nodesites/

NETWORK MAP

The Internet2 infrastructure platform delivers trusted performance, resiliency, security, and fully automated capabilities demanded by advanced research and education collaborations. It creates global access through connections with 100+ countries across 70 national R&E networks.

- I2 Network
  - [x] Optical Sites
  - [ ] Packet Sites
  - [ ] Cloud Services
- NA–REX

Optical Sites

The Internet2 optical system supports high-speed, scalable connectivity between campuses, cloud providers, computing sites, and scientific instruments. Based on the open line system concept and flex-grid technology, the system currently hosts 1.6 Tbps wavelengths while supporting community spectrum and network overlays.

- Fiber Infrastructure
- Infrastructure Sharing
- Leased Capacity



https://internet2.edu/infrastructure/i2-network-map/

Plan: Networking module



Part I. Internet protocols …and attacks!

- Local-area networking: Ethernet, ARP, DHCP, …
- Internet protocols: TCP and UDP
- DNS
- Case study: Mirai Botnet

Part II. Cryptographic protocols …and attacks!

- Transport-layer security (TLS)
- Encrypted messaging: Signal protocol (WhatsApp, iMessage, etc.)

Part III. "Bonus" lecture – Topic by popular demand

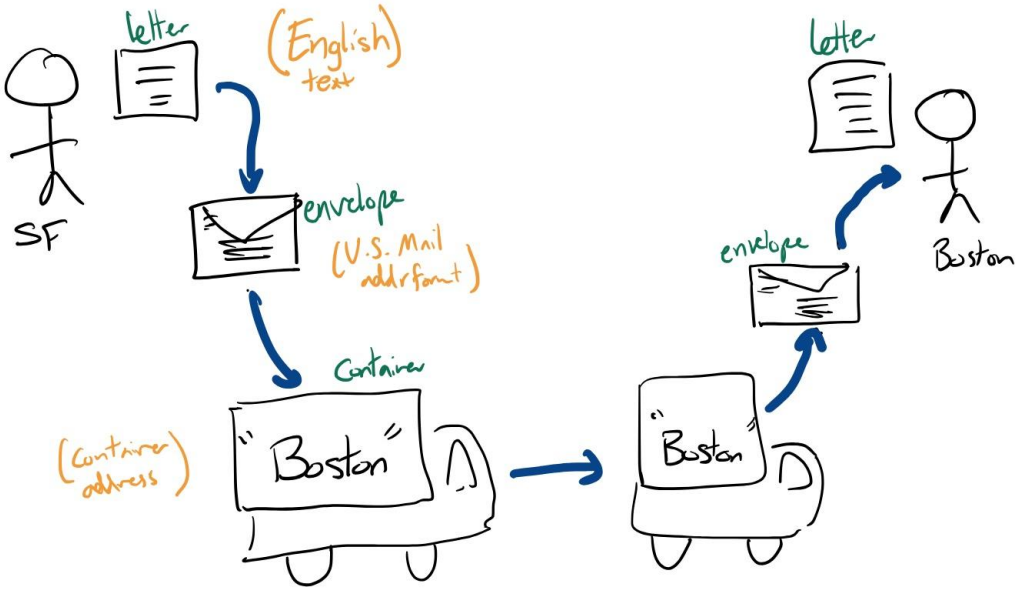
→ Submit at henrycg.com/feedback

Layering by Analogy: U.S. Mail

One global network: Mail system

There are different “protocols” for shipping:

- Letters
- Boxes
- Live animals
- …

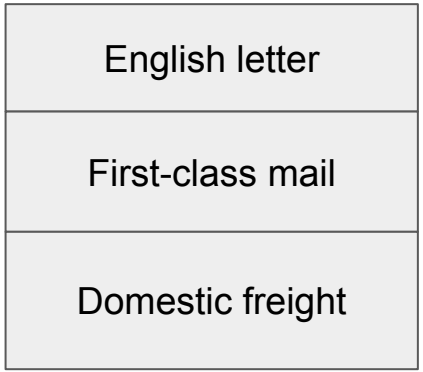


Protocols

- Each layer has a “protocol” that imposes requirements and give guarantees

- **Requirements**
  - First-class mail
    - Envelope has to be > 3.5” x 5”
    - Must have delivery address on one side
  - Domestic freight
    - Filled truck container must weigh < 30,000 kg

- **Guarantees**
  - First-class mail
    - Delivered in ≤ 5 days
    - Will contain postmark with date of processing
  - Domestic freight
    - Container will be delivered unopened



Benefits of layering

- Separation of concerns
  - When you are mailing a letter, you don’t need to care if it gets there by boat/train/truck
  - Post office doesn’t need to know if your letter is written in English or French or Spanish
- Economies of scale
  - We only need a few sizes of freight truck—all containers are interchangeable
  - Makes it much cheaper to build containers and trucks
- Enables innovation
  - Can change implementation of lower layer without having to change upper layers
  - Can invent new higher-layer protocols without changing lower layers
    - E.g., you can write your letter in a new made-up language if you want

Drawbacks of layering

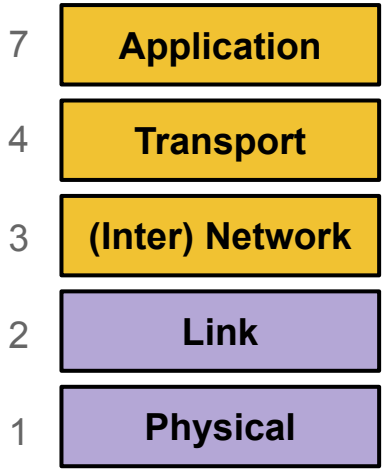
- Lower layers might not offer features you want
  - All first-class letters ship with the same speed, even the important ones
- Lower layers are “dumb” – not aware of what’s going on above
  - ⇒ Possibly inefficient
  - If I send you 10 letters on the same day, they’re treated as 10 independent objects
  - If someone mails you an empty envelope, it will still get delivered
- Hard to change protocol later on
  - Good luck changing size requirements for first-class mail

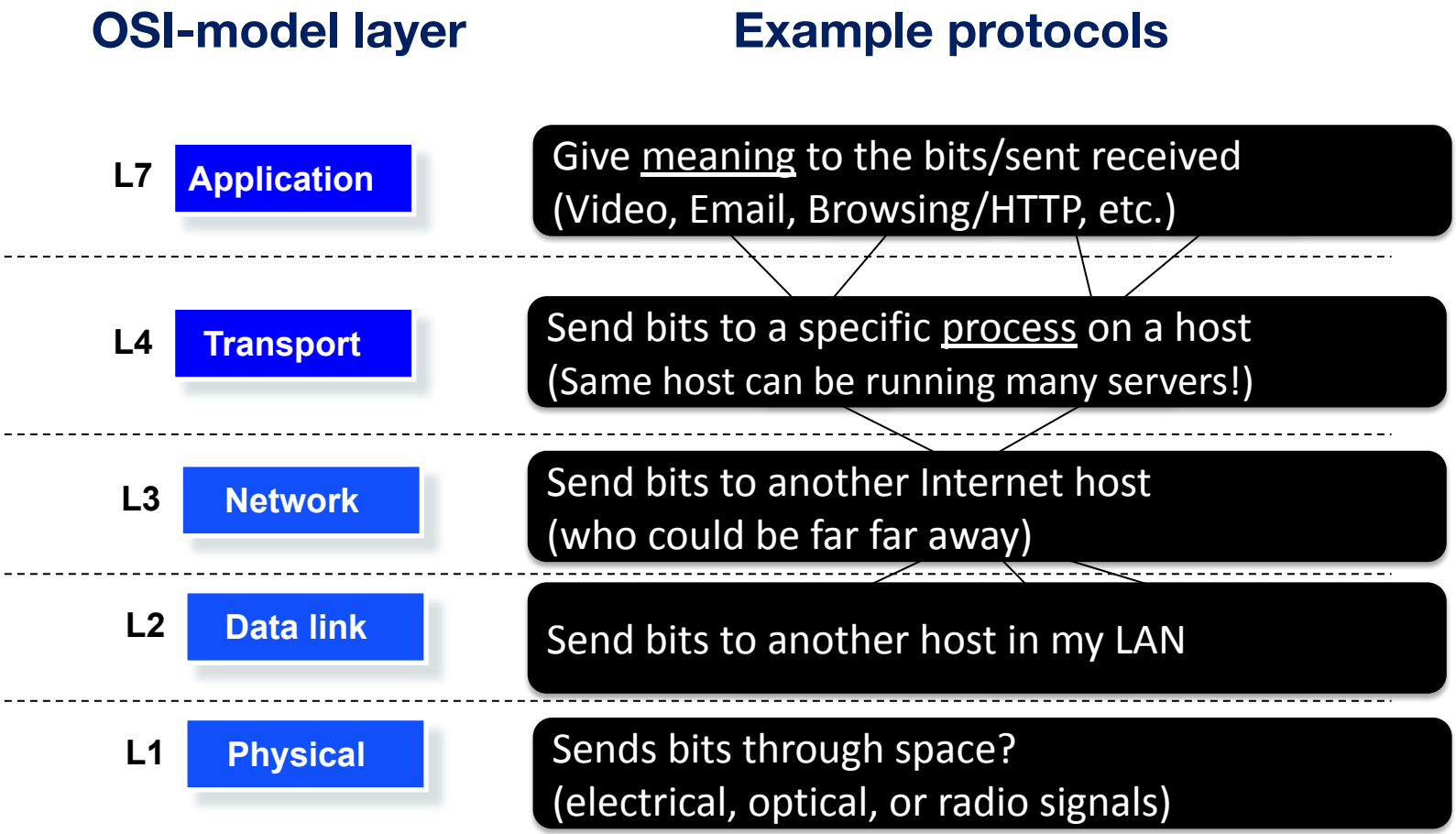
Layering: The OSI Model

OSI Model

- **OSI model:** Open Systems Interconnection model, a layered model of Internet communication
  - Originally divided into 7 layers
    - But layers 5 and 6 aren't used in the real world, so we ignore them

- As in our mail example, each layer imposes requirements and makes guarantees
  - A layer can be implemented in different ways without affecting other layers
  - A layer's protocol can be substituted with another protocol without affecting other layers





Slide courtesy of Grant Ho (UChicago) + my edits

OSI-model layer		Example address
L7	Application	HTTP: `/files/logo.png`
L4	Transport	TCP/UDP port: `80`
L3	Network	IP: `141.193.213.10`
L2	Data link	Ethernet: `36:7c:86:5c:6d:40`
L1	Physical	

Slide courtesy of Grant Ho (UChicago) + my edits

Show `ifconfig`

Back to our cartoon version of the Internet

